NORMS OF PULLBACKS ALONG T LAND IN THE INVARIANTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

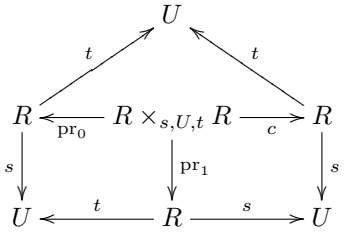
- Lemma lemma-diagram (canonical tag 02YE). cited at proof lines 4479-4480. The commutative diagram; its top part gives pr_0 t = c t.
- Equation equation-invariants (03BF). cited at statement line 4473. The invariant ring C.

Source reference. Original equation 03BF: sources/stacks-chow/tags.txt, source lines 4346–4346. Cited at statement lines 4472-4473 fixing C; navigation for the setup.

Source reference. Original lemma 02YE: sources/stacks-chow/tags.txt, source lines 3873–3873. Cited at proof lines 4479-4480 for the commutative diagram whose top part gives the equality used; load-bearing.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(A)$ and $R = \mathrm{Spec}(B)$ are affine and $s, t : R \to U$ finite locally free. Let C be as in (03BF). Let $f \in A$. Then $\mathrm{Norm}_{s^\sharp}(t^\sharp(f)) \in C$.*

Proof. Consider the commutative diagram



of Lemma 02YE. Think of $f \in \Gamma(U, \mathcal{O}_U)$. The commutativity of the top part of the diagram shows that $\mathrm{pr}_0^\sharp(t^\sharp(f)) = c^\sharp(t^\sharp(f))$ as elements of $\Gamma(R \times_{S,U,t} R, \mathcal{O})$. Looking at the right lower cartesian square the compatibility of the norm construction with base change shows that $s^\sharp(\mathrm{Norm}_{s^\sharp}(t^\sharp(f))) = \mathrm{Norm}_{\mathrm{pr}_1^\sharp}(c^\sharp(t^\sharp(f)))$. Similarly we get

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$t^\sharp(\mathrm{Norm}_{s^\sharp}(t^\sharp(f))) = \mathrm{Norm}_{\mathrm{pr}_1^\sharp}(\mathrm{pr}_0^\sharp(t^\sharp(f)))$. Hence by the first equality of this proof we see that $s^\sharp(\mathrm{Norm}_{s^\sharp}(t^\sharp(f))) = t^\sharp(\mathrm{Norm}_{s^\sharp}(t^\sharp(f)))$ as desired. $\square$